Computer Vision Team Report

Our two tasks, what we must prove, and what we need.

45	2	1
points from CV	tasks	attempt each

Our two tasks

Task 1: Count the invasive crabs — 15 pts

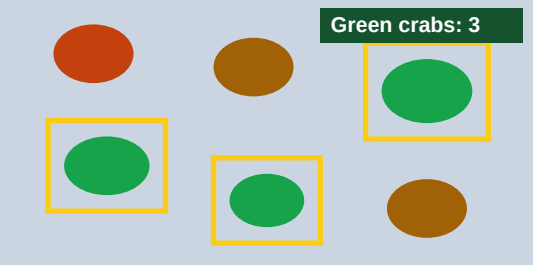
In one line: the program must find the European Green crabs on a sheet and show how many there are.

What the ROV sees

- A 50 x 50 cm sheet with printed crab pictures.
- Three kinds of crab: **European Green** (the one we want), **Rock** and **Jonah** (ignore them).
- Pictures can be bigger, smaller or rotated.

What the program must do

- Draw a box around every green crab only.
- Show the total number on the screen (7 crabs means "7").



Example: boxes only on green crabs, and the count on screen.

Success: correct count shown by the program, with the whole sheet visible to the judge.

Watch out: only one chance to report the count. People cannot count or guess. No boxes on Rock or Jonah crabs.

Our plan: train a detector on all 3 crab types, with many rotations and sizes, and count over several frames for a stable number.

Task 2: Move the coral by itself

30 pts

In one line: after the ROV picks up the brain coral, it must drive itself to the red square and put the coral down.

1. Pick up	2. Leave nursery	3. Travel	4. Find red square	5. Place coral
Pilot, manual	Handoff point	Autonomous	Camera	Autonomous

Orange = pilot, manual. Teal = autonomous.

The setup

- The coral starts in the nursery area.
- The goal is a 15 cm red Velcro square in the restoration area.

The rules

- After the coral leaves the nursery, nobody touches the controls.
- The tether manager can hold the tether but cannot guide the ROV.

Success: any part of the coral's bottom touches the red square, with no manual control after pickup.

Watch out: one attempt only. If the pilot takes control after pickup, we lose all 30 points.

Our plan: find the red square in the camera image, steer toward it, hold pitch with the IMU, and release the coral on top.

Vision requirements

Requirement	In simple words	Applies to
Real time	Fast enough to work live	Both
Accuracy	Good enough to be reliable	Both
Autonomous motion	One activation, then no user input	Coral
Dataset	Public or our own	Both
Deployment	Runs on the real ROV	Both
Robustness	Works in any position, rotation, lighting	Both
Testing	Test data never seen in training or tuning	Both

Careful: the crab task uses only 3 source photos. Make the test set from new real captures, not copies of the training images.

What we need

From the software team	Other needs
Camera feed and IMU over ROS 2 **Autonomous mode in the GUI** **Agreed topic names**	GPU for training Printed crab sheets, red square mockup Regular pool test time

Main risks

- Red looks dark underwater: calibrate in the pool.
- Green and Rock crabs look alike: add hard examples.
- One chance: count over several frames.
- Slow model: pick a small one early.

Deliverables

- GitHub repo: training, tuning, deployment, autonomous code
- Dataset link
- Demo on the real ROV
- Presentation: features, architecture, design choices

Based on the Final Project manual, Season 26-27.